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Subject Code:MPC204T

Roll No:

MPHARM**(SEM. II) THEORY EXAMINATION 2025-26****PHARMACEUTICAL PROCESS CHEMISTRY****TIME: 3 HRS****M.MARKS: 75****Note: 1.** Attempt all Sections. If require any missing data; then choose suitably.**SECTION A****1. Attempt *all* questions in brief.****10 x 2 = 20**

a.	Define process chemistry.
b.	What is azeotropic distillation?
c.	Write any four method of extraction.
d.	What is Ozonolysis?
e.	Difference between azeotropic and steam distillation.
f.	Discuss the types of fire hazard.
g.	Write various sources of impurities.
h.	Define catalytic hydrogenation.
i.	What is nitration?
j.	What is PPE?

SECTION B**2. Attempt any *two* parts of the following:****2 x 10 = 20**

a.	Explain stages of scale-up process (bench, pilot, industrial scale).
b.	Define impurities of API. Explain their types with suitable examples.
c.	Discuss aromatic nitration mechanism and industrial aspects.

SECTION C**3. Attempt any *five* parts of the following:****7 x 5 = 35**

a.	Differentiate between Steam Distillation and Azeotropic Distillation.
b.	Explain in process control and validation of large- scale process.
c.	Discuss preparation of hydrates, solvates and amorphous forms.
d.	Discuss in detail about material safety data sheet (MSDS) ISO.
e.	Discuss in detail about 14001(Environmental management System).
f.	Discuss production of antibiotics (Penicillin/ Streptomycin).
g.	Discuss fire hazards and safety management systems (OHSAS, ISO).